Theft & Stalking: An Evaluation of Current Bluetooth Low Energy Tracking Device Capabilities

Salem Radey
UW–Madison

ABSTRACT

Modern consumer Bluetooth low energy (BLE) tracking devices are designed to help users locate items and many users additionally use these devices to recover items after theft. However, due to the nature of these tracking devices, they can also be used to facilitate stalking and intimate partner violence by allowing a user to track another person without their knowledge. To counter this, many manufacturers have added 'anti-stalking' features such as proximity alerts, audio pings, and in-app detection scans.

In 2023 researchers at Apple and Google submitted the first draft of the Detecting Unwanted Location Trackers (DULT) charter to the Internet Engineering Task Force (IETF) with the goal of standardizing the construction and behaviors of BLE tracking devices to allow the issuance of unwanted location tracking alerts. Now, in 2025, this charter has been approved by the IETF.

Despite the push by industry leaders for stronger anti-stalking protections, many users and manufacturers worry that the stringent guidelines may make it easier for a thief to notice a tracking device; thereby rendering a perceived core functionality of the device obsolete.

This paper seeks to quantify the trade-offs of anti-theft and anti-stalking made by popular devices and apps based on their policy. We compare the written policy of Life360 Tile Mate (2024), Apple AirTag, Samsung SmartTag2, Chipolo ONE, Pebblebee Clip 5, and their companion apps. Additionally, this work investigates manufacturer adherence to DULT protocols as laid out in the IETF charter. We find that across 111 DULT guidelines, adherence rates ranged from 5.4% - 21.6% with Apple AirTags having the highest adherence rate and Life360 Tile Mate (2024) trackers having the lowest adherence with the most amount of explicit violations.

1 INTRODUCTION

Originally created by Nokia under the name "Wibree" in 2006 [18], Bluetooth Low Energy (BLE) technology was integrated into Bluetooth 4.0 in 2009 [21]. Since its inception almost 20 years ago, BLE has gone on to facilitate the creation of a wide range of technologies [10]. One such use case for BLE is tracking devices sometimes referred to as "smart tags". By sending packets to a paired device which can then send location data to a server, a smart tag is able to be found by its owner using a corresponding app.

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By affixing a smart tag to an item, a user can find the item by consulting the tag's location with the corresponding app. Although these smart tags make finding misplaced items easy, they also can be misused by attackers to facilitate stalking.

To mitigate the potential for misuse, researchers at Apple and Google worked to create the Internet Engineering Task Force (IETF) charter, "Detecting Unwanted Location Trackers" (DULT) [24]. DULT enumerates various requirements and recommendations for Bluetooth tracking device manufacturers to adhere to. For example,

> The accessory SHALL maintain an internal state that detects when its location is, or has been, available to the owner via a network ("location-enabled state")

The creators claim that the goal of the DULT charter is to create a set of best practices which will enable compatibility with unwanted tracking alert detection on phones and help preserve user privacy and safety [24].

Research has been done which investigates the various ways modern smart tags can be abused by adversaries to execute stalking attacks and perpetuate intimate partner violence [11]. Additionally, past works have reverse engineered popular BLE trackers such as the Apple AirTag and the Life360 Tile trackers to find various vulnerabilities [22, 40].

Since many studies only focus on a few devices, there remains a gap in understanding of what attacks are possible per manufacturer on a wider scale as well as a policy-level analysis of the claims manufacturers are making. Additionally, DULT was proposed a few years ago, yet there are no studies which seek to empirically evaluate the current adherence rate of existing manufacturers.

This paper makes three contributions: (1) a structured policy-level analysis of five major commercial BLE trackers, (2) a cross-manufacturer comparison of attack surfaces under two threat models, and (3) the first systematic analysis of DULT adherence.

The following research questions are used to motivate these contributions:

(1) **RQ1:** What attacks are possible for each device under the theft threat model?
(2) **RQ2:** What attacks are possible for each device under the stalking threat model?
(3) **RQ3:** Although we have created DULT protocols, are they being adopted by the manufacturers?

To evaluate these research questions, we focus on a written policy-level analysis of Life360 Tile Mate (2024) [29], Apple AirTag [4], Samsung SmartTag2 [37], Chipolo ONE [15], and Pebblebee Clip 5 [34]. To answer **RQ1** and **RQ2**, we collect these policies by querying Google with various targeted terms. The top 3 results returned are noted, read, and annotated. From there, we additionally note and read the webpages which are linked from within the queried results. Using the annotated webpages, we then enumerate possible

attack vectors for each manufacturer under two separate threat models: the theft threat model and the stalking threat model.

To answer **RQ3** we systematically evaluate the adherence of each of the 5 devices to 111 DULT guidelines which contain the words, "MUST", "MUST NOT", "SHALL, "SHALL NOT", and "REQUIRE". Each requirement is given a "YES", "NO", or "NO EVIDENCE" rating based on the information found from the webpage annotation stage.

We find that Apple AirTags adhere to DULT guidelines the best, with 24/111 requirements met based on policy-analysis. On the other hand, Life360 Tile Mate was found to adhere to DULT the least with only 6/111 and has the highest number of "NO"'s compared to other devices.

2 BACKGROUND

Before discussing related works and our findings, we will first provide necessary background information on BLE, tracking device offline finding networks, and DULT best practices. Additionally, we clarify the two threat models. Throughout this work, "tracker", "tracking device", "tracking tag", "smart tag", and "tag" will be used interchangeably.

2.1 Bluetooth Low Energy

Bluetooth Low Energy technology was integrated into Bluetooth 4.0 in 2009, touting far lower power consumption than prior Bluetooth instantiations. BLE operates with a unique, independent, protocol distinct from classic Bluetooth.

BLE allows for small bursts of data to be transmitted over radio frequency (specifically on the 2.4 GHz ISM band) over a short range. This contrasts classic Bluetooth which can be used for larger amounts of data transmission, such as data streaming. [23] BLE does this with a small enough power consumption to allow devices functioning on a limited amount of power such as button batteries to keep charge for much longer.

Tracking tags operate using this BLE protocol. They have only small amounts of power, low energy and compute resources, and are unable to connect to the internet. Within short ranges (around 60-150 meters for devices in this study) [30, 34, 39] the tags are able to transmit Bluetooth beacon messages which can be received by a device listening on the frequency. The tracking tags are associated to an owner device (for example, an Apple AirTag may be will be associated to an Iphone as its owner device) which allows a user to interact with the tag through a paired app. Interactions may include functionality such as making the tracker emit a sound or flash a light to facilitate finding the tag. If the companion app on the owner device does not detect these advertisements for an associated tracker, it may issue an "out of range" alert. Out of range alerts are notifications which notify a user that their tracking device is possibly out of the Bluetooth range and may have been left behind. If a tag cannot communicate to an owner device due to being out of range, it may utilize an offline finding network.

2.2 Offline Finding Networks

Offline finding networks allow for tracking tags to be located even when outside of Bluetooth range to their owner associated device. BLE beacon messages sent by the tracking tag include a rotating ephemeral public key which can be received by nearby bystander devices. [2] Bystander devices are devices in the tag's "finding network" and what devices this encompasses varies by manufacturer. For Example, Apple AirTags can leverage bystander devices which participate in the Apple Find My finding network include IOS devices with Find My enabled (which is enabled by default on Iphones). The bystander device will then encrypt the location with the tag's public key. These reports are uploaded to a server which can be queried by the tag owner's associated device to decrypt with their corresponding private key and retrieve the approximate location of their tracking tag. [2]

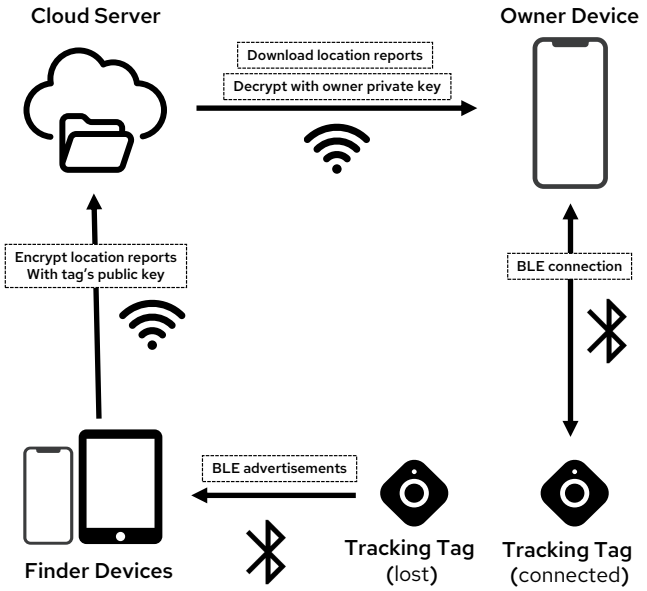


Figure 1: Depiction of a tracking tag's connection to an associated owner device and offline finding network. Adapted from [22]

2.3 DULT

Due to the sensitive nature of location data, adversaries can use Bluetooth location tracking accessories such as tracking tags to commit acts of stalking, theft, and harassment. To mitigate this threat, researchers at Apple and Google have worked together to submit a charter to the Internet Engineering Task Force (IETF) known as the Detecting Unwanted Location Trackers (DULT) protocol. The protocol enumerates a list of best practices containing the key terms "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" which specify the degree to which a manufacturer needs to adhere to the specific practice. [24] For the sake of this study, we look exclusively at the listings which require adherence falling under the key terms "MUST", "MUST NOT", "REQUIRED", "SHALL", and "SHALL NOT".

2.4 Threat Models

We will consider two threat models; one for stalking and one for theft.

(1) The **theft threat model** assumes that an attacker will be able to physically take a victim's property which is affixed with some tracking device. Initially, we assume the attacker does not know of the tracking device.
(2) The **stalking threat model** assumes an attacker will place or access a tracking device on the victim / the victim's belongings. Initially, we assume the victim does not know of the tracking device.

3 RELATED WORK

Research into the safety, privacy, security, and functionality of BLE trackers remains a growing domain. We separate the related works into the following categories: safety alerts, security and privacy, location accuracy, and research done in other fields.

3.1 Safety Alerts

Safety alerts allow a device to alert a user when an unassociated tracking device may be traveling with them. A large motivation for the creation of DULT best practices was to allow for safety alerts to successfully identify trackers across manufacturer lines. To this end, various studies have investigated what triggers these alerts as well as their accuracy [17, 40].

The work done by Shafqat et. al. investigated Apple AirTag's safety alerts to try to discern what triggers them as well as the responsiveness of said triggers. They found that alerts can take over 8 hours to appear during the day. Furthermore, they demonstrated the ability to circumvent safety alerts by creating a spoofed AirTag which works directly on the Find My app.

Similarly, Gerhardt et. al. also evaluated the reliability of AirTag safety alerts and found them to be much more responsive on Apple devices than Android. Safety alerts are a strong step towards stalking mitigation, however these works demonstrate that they alone are not sufficient to offset adversarial uses of tracking devices.

3.2 Tracker Security and Privacy

Another possible way to exploit tracking devices lies within their cryptographic schemes and transmission protocols. Prior research investigates the various means trackers can be exploited through such as insufficient cryptography, passive eavesdropping, physical tampering, and more [1, 22, 42].

Alamleh et. al. investigated Apple AirTag and Samsung SmartTag trackers and found various vulnerabilities through multiple vectors such as, "physical tampering, firmware exploitation, signal spoofing, eavesdropping, jamming, app security flaws, Bluetooth security weaknesses, location spoofing, threats to owner devices, and cloud-related vulnerabilities" [1].

Yu et. al. also chose to focus on SmartTags, and likewise found a plethora of ways to persistently track a user leveraging the offline finding network. Yu calls attention to the current fragmentation of the tracking device ecosystem and mentioned hope that the DULT protocol will have a positive change on the industry and help to mitigate some of the aforementioned vulnerabilities [42].

Although many studies focus on AirTags or SmartTags, Kumar et. al. helped to close the research gap by conducting the first comprehensive study to reverse engineer Tile trackers. Likewise to other trackers, they find various vulnerabilities which allow the platform itself to persistently track its users - explicitly disproving many of the company's privacy claims. Despite the value in showing systems-level vulnerabilities of these trackers, all of these studies limit the scope to one or two trackers which makes it difficult to make a comparison of the security guarantees and risks across manufacturers.

3.3 Location Accuracy

For location tracking devices it naturally follows that location accuracy would be a primary vector to gauge the effectiveness of a device both for users and adversaries.

In 2023, Ibrahim et. al. investigated the accuracy of Apple AirTag and Samsung SmartTag trackers in both controlled and in-the-wild experiments [19]. They found that the two devices performed similarly, and a victim's movements can be accurately backtracked with just an hour's delay.

Then in 2025, Jang et. al. conducted a study which extended the original 2023 study [20]. In addition to AirTags and SmartTags the study added Tile trackers. Continuing with controlled and in-the-wild experiments the authors were able to note that AirTags had increased their location reporting frequency. Pertinently, they determined that the primary factor for location accuracy had nothing to do with device specification, but rather the extensiveness of the finding network.

This suggests that manufacturers with a larger finding network such as Apple Find My have an accuracy advantage over other devices with tout a smaller network such as Chipolo, although an extensive study across a wide range of manufacturers has yet to corroborate this outside of Jang's study which was limited to three devices.

3.4 External Fields

The threat of misused trackers has spurred research on the topic in other fields such as law and digital forensics [11, 41].

Bonetti's work displays the way in which Apple AirTags effect vulnerable populations. They highlight how sex trafficking, domestic violence, theft, and stalking can be perpetrated with tracking devices. They illustrate how the digital age is changing our definition of stalking, as well as how legislation has not caught up to this change which makes persecution difficult [11]. In that same vein, progress in the digital forensics field has been made by Yang et. al. who introduce S.TASER (Smart Tag Parser) which allows for the identification of deleted tags and enables the recovery of identification data. [41]. Although they approach this from a forensics lens, we are left to wonder if this the ability to retrieve said data could be leveraged by an attacker.

3.5 The Gap

Looking at these prior works, a research gap emerges. Many studies typically investigate one, sometimes two, and rarely three, tracking tag manufacturers. It is clear there are various attack vectors against BLE tags, but a quantifiable survey of these vectors is not present. Additionally, multiple recent studies cite DULT as a possible remedy for many adversarial concerns, yet there has not been a survey of how many manufacturers are adopting DULT and if adopting DULT results in less attack vectors.

4 METHODOLOGY

The methodology consisted of four distinct phases: The web query collection phase, the web query annotation phase, the attack vector formulation phase, and finally the the DULT adherence phase. Phases 1, 2, and 3 correspond to RQ1 and RQ2 whereas phase 4 correlates to RQ3.

4.1 Web Query Collection

To conduct a policy-level analysis of the devices the published policies of the various manufacturers and devices are collected using a manual, structured, web query collection. Web queries are chosen as the basis for collecting official policy since they serve as primary sources from the manufacturer. Additionally, firmware and hardware level schematics often are closed-source. Therefore, external policy is the only verifiable source without conducting reverse engineering since manufacturer privacy policies may explicitly disallow [13].

First, a limited list of queries is formulated. These queries follow the format of [QUERY] [MANUFACTURER] [DEVICE] where the manufacturer list is Life360, Apple, Samsung, Chipolo, and Pebblebee. Corresponding to the list of manufacturers, the list of device keywords is Tile, AirTag, Galaxy SmartTag, Chipolo One, and Clip 5. A list of these queries can be found in table 1.

For each query, the top 3 results from official sources are noted. For example, for Apple AirTag this means only results from apple.com and its subdomains are noted. This ensures that the links are in fact official policy and not speculation. For this reason, prewritten help articles are also included but forums are not (even official ones, such as discussions.apple.com) since those are support employee responses and do not necessarily reflect official policy.

Some queries were unable to produce 3 results from official sources within a reasonable amount of pages. We consider a reasonable amount of pages to be ten Google results pages. Additionally, some devices struggled with SEO semantics which caused erroneous pages to be returned instead of more pertinent ones. One common example of this was with Chipolo ONE. Chipolo also has devices named Chipolo ONE Spot and Chipolo ONE Point alongside the Chipolo ONE which is included in this study. Chipolo ONE works off of Chipolo's proprietary finding network whereas the Spot and Point work off of Apple Find My and Google Find My Device networks respectively. Therefore, many results returned under the Chipolo ONE queries were inapplicable.

Overall, 45 unique links are identified from 4 queries which each take the top 3 results for 5 manufacturers. Some links were duplicated across multiple queries, which explains the lower than expected number (60) of results. A full list of collected links per query can be found in the appendix section 8.1.

links which were hyperlinked from within the queried results were also accessed to get a more complete view of the possible policies noted online. Additionally, a few unique queries were necessary later in the process for DULT adherence evaluation (ex: "how to disable Chipolo ONE"). After removing inapplicable links and adding links from snowballing and unique queries 44 links are left to be referenced. This list can be found in the appendix section 8.2.

Table 1: Full list of queries

List of Queries
Anti-stalking [manufacturer] [device]
Unwanted tracking [manufacturer] [device]
Anti-theft [manufacturer] [device]
Privacy policy [manufacturer] [device]

4.2 Web Query Annotation

To effectively and accurately reference the written policy for the attack formulation stage and DULT adherence evaluation the queries must be annotated. This annotation process involves reading each policy and documenting pertinent information by assigning sentences a category label and saving the line as a quote or summary for future reference. Direct quotes are used when the referenced line contains important nuance, whereas summaries are used when the referenced line is simple in nature. Pertinent information is considered to be any reference to technical specifications, references to unwanted tracking or theft, and references to features which could be exploited by an attacker. Category labels include "anti-theft mode", "anti-stalking", "privacy policy", "DULT", "basic functionality", "device specifications", and "anti-harassment". For example, a page may have an entry of "Life 360 Tile Mate / Anti-Theft Mode / 'If your verification fails, you can try again' / [link]". Each page may have multiple annotation entries.

All annotations capture four fields: (1) vendor and device, (2) assigned category, (3) extracted statement (quote or summary), and (4) the source link. These structured annotations form the basis for identifying claimed protections, potential misuse conditions, and DULT-relevant behaviors analyzed in later sections.

4.3 Attack Vector Formulation

Using the annotated policies from the previous section, we construct possible attack vectors for each device under both the stalking and theft threat models. Since this work focuses on policy-level analysis rather than systems level data, the attack surface is analytically constructed. The attacks proposed represent behaviors an attacker could leverage based on the capabilities and limitations demonstrated within the collected policies as noted by each manufacturers publicly available documentation.

For each tracker we provide a minimum of one attack scenario for each threat model. The Life360 Tile Mate (2024) tracker is split into two distinct operational modalities: "regular" mode and "anti-theft" mode. Tile Mate trackers documented behavior and security properties differ significantly once in Tile's Anti-Theft mode which requires additional examination. The complete attack descriptions and their security evaluations appear in Section 5.1.

4.4 DULT Adherence Checklist

To evaluate how closely each vendor's documented behavior aligns with the Device Unwanted Location Tracking (DULT) specification, we construct a structured DULT adherence checklist.

DULT requirements are translated into a series of trinary checklist items (e.g., supports cross-platform alerts, rotating BLE identifiers, owner-device proximity-based suppression, clear unwanted-tracking disclosure, anti-stalking alert timing). Each checklist item

is framed such that the manufacturer's public documentation either explicitly claims support for the requirement, contradicts it, or provides no verifiable statement. Because this study is limited to policy-level analysis, adherence is evaluated strictly on the basis of publicly stated guarantees rather than implementation-level verification. Only DULT requirements which contain the coded keywords "MUST", "MUST NOT", "SHALL, "SHALL NOT", and "REQUIRE" are factored into the list of 111 guidelines. Our complete DULT checklist can be found in Section 8.3. Note that the table has more than 111 requirements due to the listings "features NFC", and "features accelerometer" being added by the author and not from the original checklist. This was added for brevity since if a tracker does not feature NFC or an accelerometer, it does not need an entry for the subsequent guidelines specific to said functionality.

For each device, every checklist item is assigned one of three outcomes:

- Yes: documentation explicitly states adherence to the requirement.
- No: documentation explicitly states the device does not adhere to the requirement, or presents a design choice incompatible with the requirement.
- No evidence: documentation provides insufficient or ambiguous information to assess adherence.

5 EVALUATION

This section will cover the formulation and analysis of possible attack vectors per manufacturer, as well as analyze the results from the DULT adherence checklist.

5.1 Attack Vectors

Using the annotated policy data described in Section 4 we evaluate feasible attack vectors under both the theft and stalking threat models (Section 2.4). Each attack leverages features explicitly described by the manufacturer or behaviors implied by the absence of disclosed safeguards.

5.1.1 Life360 Tile Mate (2024) - Normal Mode. Tile trackers operate off of the Life360/Tile offline finding network. This network consists of all devices which have the Tile or Life360 apps downloaded [26, 31]. Tile features a Scan and Secure mode which allows a user to actively check for nearby Tile trackers. This feature only detects Tile brand devices and no other tracker. Additionally, this feature takes approximately 10 minutes to execute [32].

Theft Threat Model

- Tile's Scan and Secure feature allows users to actively scan for nearby Tile devices that may be "traveling with you" [32]. Under the theft threat model, an attacker who steals an item affixed with a Tile Mate may identify the tracker by performing an active scan and subsequently discard or disable the tag. However, this attack requires the attacker to both suspect the presence of a Tile device and and deliberately initiate a scan using the Tile application. A savvy attacker may consider this, however it does make the attack harder to execute.

Stalking Threat Model

- Similarly to the theft threat model attack, since Tile does not support continuous or passive detection of unknown trackers and detection requires a time-intensive active scan lasting approximately 10 minutes a victim who does not suspect a tracker (particularly a Tile device) may remain unaware of its presence indefinitely. Additionally, based on available policy documentation, it is unclear whether Tile devices are consistently detectable by Apple's Find My or Google's Find My Device unwanted tracking alert systems. Tile makes no claims of being DULT compliant and Apple notes that devices must be compatible with DULT specifications to issue unwanted tracking alerts [6].
- Further, Tile's Share a Tile feature allows an owner to authorize additional users to ring the device, view its location, and receive notifications [27]. In an intimate partner setting, an attacker could exploit this feature by covertly adding themselves as an authorized user to a victim-owned Tile attached to frequently carried items (e.g., keys). Because the tracker remains legitimately owned, this attack would not be detectable via Tile's Scan and Secure mechanism. To remove the shared person, a victim would need to go into the Tile app, select the specific tracker, select "Unlimited Sharing" and then remove the email associated with the shared person. This feature is nested within a few menus and gives no reminder that your Tile Mate is being shared. Therefore, it is likely to assume that a victim may not notice. This feature is popular across many devices and therefore is an attack vector for many other trackers in this study. This attack will henceforth be referred to as the "sharing attack".

5.1.2 Life360 Tile Mate (2024) - Anti-Theft Mode. Tile trackers also have a unique feature known as Anti-Theft Mode. This Anti-Theft mode enables a Tile tracker to not appear in the Scan-and-Secure feature [25]. To be eligible to activate this feature users must first upload a front and back photo of an ID. Then, users are required to do a live 'selfie-scan' [28]. Tile also states that users may try to verify an ID 3 times, and if it fails they should attempt a different ID [28].

Due to this added functionality, we give a Tile Mate (2024) in Anti-Theft mode separate consideration with regards to its threat capabilities.

Theft Threat Model

- Tile's anti-theft mode means an attacker will not be able to actively scan for the device, even with the Scan-and-Secure feature. However, like all Bluetooth trackers which rely on an offline finding network, it is possible the attacker could go 'off grid' to somewhere that the tile network cannot track. This is unlikely given the scope of Tile's network. Otherwise, the attacker could attempt to put the entire stolen item in a Faraday cage or wrap it in tinfoil to stop the signal transmission [32]. Since Tile Mate (2024) does not contain a removable battery, those options are the only way to stop it from transmitting location data. [30] Though, this would only work until the attacker needed to access the item.

Stalking Threat Model

- When Anti-Theft mode is enabled, the Tile tracker will not show up in tile's scan-and-secure feature. Therefore, an attacker could easily turn on Anti-Theft mode and slip the tracker onto the victim's person. As previously stated, Tile trackers might not issue unwanted tracking alerts since they do not claim to be DULT compliant and only compatible trackers issue passive alerts. Therefore, with the inability to actively scan using Scan-and-Secure mode and the inability to receive passive alerts a victim will have no way to know they are being tracked. Anti-Theft mode makes a stalking attack trivial.
- Similarly to the regular mode, the sharing attack is still possible while in Anti-Theft mode.

5.1.3 Apple AirTag. Apple AirTags operate using the Apple Find My network. A vast majority of iOS devices act as finder devices in this network since the feature is turned on by default for all devices operating iOS 13 or later, iPadOS 13.1 or later, and macOS 10.15 or later [3]. This gives the Find My network the largest finding network of all devices in this study, consisting of hundreds of millions of finder devices [5].

Theft Threat Model

- To prevent tracking abuse, AirTags emit a sound whenever moved if the AirTag is not associated with its owner [9]. Therefore, if an attacker steals an item with an AirTag it will begin to emit sound when the attacker moves. Then, the attacker can simply disable the AirTag by removing the battery [9] or disposing of the tracker. This makes a theft attack quite simple.

Stalking Threat Model

- A stalking threat model attack is difficult to execute using an Apple AirTag. There are many functionalities in place to alert a victim to its presence. Upon sensing movement using its accelerometer the AirTag will beep to alert those around it to its presence. Additionally, AirTag will send an Unwanted Tracking alert on iphone and android phones. An attacker would need to alter the AirTag to remove the sound emitter. However, that would not stop the unwanted tracking alert.
- An attack which could be executed is the sharing attack. Apple AirTags can be shared with another user (called a borrower) by entering their Apple account information into the FindMy app. This will issue an invitation which the borrower can accept. Apple states that borrowers will not receive unwanted tracking alerts when traveling with the device and you will be able to locate the AirTag with precision finding due to the AirTag's Ultra Wideband technology [7, 8]. This could be used by an attacker in an IPV setting to weaponize a victim's tracker against them.

5.1.4 Samsung Galaxy SmartTag2. Like Tile trackers, SmartTags operates of a own proprietary finding network: Samsung SmartThings. SmartThings has an active scanning feature, but no passive scanning feature. Additionally, the SmartThings network is relatively small compared to other manufacturers which means there are less devices for the tracker to have its location data correlated to.

Theft Threat Model

- A theft attacker could locate the tracker with the help of the SmartThings active scanning features. However, the attacker would need to know to download the SmartThings app to check. Documentation also notes that the SmartThings active scanning feature may only be available to Samsung Galaxy owners, making the scanner inaccessible to other users who wish to scan for SmartTags [39]. Then, the attacker could simply discard the tracker or remove the battery.

Stalking Threat Model

- Also similar to Tile, it is unclear whether or not Apple Find My network and Google Find My Device network would be able to recognize Samsung SmartTag2's to issue a Unwanted Tracking alert. Therefore, victims might not be notified. Additionally, SmartTags are not as popular as other devices which means a victim might not think to download the SmartThings app and use the SmartThings scanner to detect the SmartTag2.
- Since SmartTag2's can be shared with other users, the sharing attack is also possible for this device [38].

5.1.5 Chipolo ONE. The Chipolo ONE line of trackers is a unique generation of trackers which includes the Chipolo ONE, Chipolo ONE Spot, and Chipolo ONE Point. The Chipolo ONE, which is focused on in this study, works off of Chipolo's own finding network consisting of devices that have the Chipolo app downloaded [12]. This is contrasted by the ONE Spot and ONE Point which leverage the Apple Find My network and Google Find My Device networks respectively.

Theft Threat Model

- Chipolo ONE trackers are not marketed as theft-recovery trackers, rather as simple devices made to help find misplaced items [15]. A stolen item affixed with a Chipolo ONE tracker would track its location until out of Bluetooth range, send and 'out of range alert' after 60 meters (if the user has them enabled), and then be unfindable by its owner. Chipolo ONE devices are unique from other trackers in this study since they do not seamlessly transfer to the offline finding network. To utilize the offline finding network, the owner needs to manually put the tracker into 'lost mode' via the Chipolo app [12]. Once in lost mode, the tracker will begin utilizing Chipolo's offline finding network.
  Due to this, a theft attacker would have a longer time to get away with the stolen item, possibly selling it or otherwise completing whatever objective they have with the stolen item. Additionally, the Chipolo app has 1M+ downloads on the Google Play Store and an unknown number of downloads on the Apple App store [14]. Although this number is not insignificant, it is much smaller than the hundreds of millions of finder devices that Apple AirTag has for example. This means their finding network is smaller and so the tracker may not broadcast to a finder device and its location may go un-updated.

Stalking Threat Model

- Traditional stalking attacks could work with Chipolo ONE trackers, but they would require the attacker to put the Chipolo in lost mode. Similarly to the concerns raised in

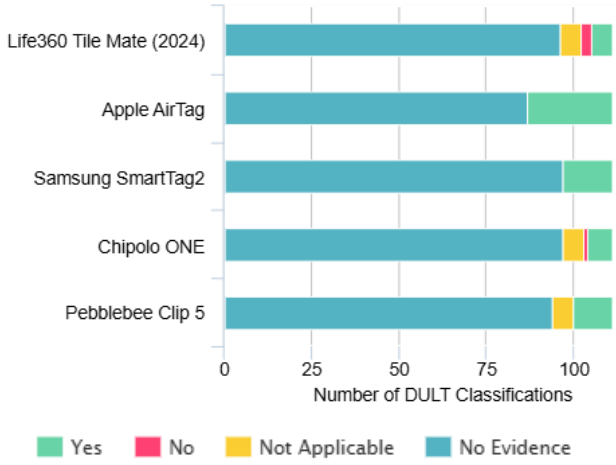


Figure 2: Data visualization of DULT findings per device.

the theft threat model, the finding network size could be a limitation for the attacker in this case.

- Chipolo ONE trackers can share their location with another person via the Chipolo app. Therefore, the sharing attack also applies here. This is done from the tracking device owner's phone by sending an invite link in the app [16]. This could be used in an intimate partner setting to surveil a partner with their own tracking device. The app does show who you are sharing your device with, but it is nested and not shown unless the victim would think to check.

5.1.6 Pebblebee Clip 5. The Pebblebee Clip 5 can operate on either the Apple Find My or Google Find My device network. As such, the attack vectors are very similar to the Apple AirTag attacks seen earlier.

Theft Threat Model

- Pebblebee devices emit unwanted tracking alerts when traveling with a user who is not their owner [35]. This would alert a thief on their phone, who would then be given explicit instructions to locate the Clip 5. The thief could then remove the battery or discard the device.

Stalking Threat Model

- Since the Pebblebee Clip 5 sends unwanted tracking alerts, a stalking attack is more difficult to execute. Additionally, Pebblebee trackers can be disabled without being reset [33]. This means a Pebblebee used to stalk may leak information about the attacker to the victim, which may deter attackers. Pebblebee also features a rechargeable, removable, battery which could be removed by the victim as well.
- The Pebblebee Clip 5 has a sharing feature, which means the sharing attack is applicable for this device as well [36]

5.2 DULT Checklist Findings

Certain DULT requirements are optional, but if they contain an optional feature there are required specifications for that feature. For example, it is not required to have NFC capabilities or an accelerometer, but if those functions exist then they have certain requirements. That is why some figures below report "not applicable". That accounts for cases where a tracker either explicitly does not have NFC capability, therefore the requirements are not applicable.

Additionally, the lines "Contains NFC" and "Contains accelerometer" are not counted in the 111 total. This is because those were not DULT requirements, but instead added for chart clarity.

The findings from the completed DULT checklist are as follows.

- Life360 Tile Mate (2024) has 6/111 requirements as "Yes", 3/111 as "No", 6/111 as "Not Applicable" and 96/111 as "No Evidence".
- Apple AirTag has 24/111 requirements as "Yes" and 87/111 as "No Evidence".
- Samsung SmartTag2 has 14/111 requirements as "Yes" and 97/111 as "No Evidence".
- Chipolo ONE has 7/111 requirements as "Yes", 1/111 as "No", 6/111 as "Not Applicable", and 97/111 as "No Evidence".
- Pebblebee Clip 5 has 11/111 requirements as "Yes", 6/111 as "Not Applicable", and 94/111 as "No Evidence".

Apple AirTags have the most amount of requirements marked as "Yes" with 24/111. Life360 Tile Mate (2024) trackers have the least marked as "Yes" with 6/111 requirements. The Tile Mate trackers have the most amount of requirements marked as "No" with 3/111.

To see a visualization of these findings, see Figure 2. To see the full DULT checklist, see Table 3 in Section 8.3.

6 DISCUSSION

6.1 Limitations

Due to this work being a policy-level evaluation, the vast majority of DULT adherence guidelines were left as "no evidence" and could not be determined conclusively. Many public facing policies are used as advertising materials and not as technical specification documents. This results in a lack of information which causes an incomplete view of the current tracking device landscape.

These limitations further influence the attack vector formulation. Because the attacks are derived from publicly claimed behaviors instead of implementation-level behavior, some vectors may be incomplete or may miss attack surfaces that only emerge through firmware analysis, BLE traffic capture, or hardware-level investigation. Conversely, some policy-described features may not exist in practice, or may behave differently from the text, causing our attack vectors to over or underestimate attacker capabilities.

Another limitation is the tracking devices chosen. At the time the study began, the Chipolo ONE line of trackers were the most popular for their brand. However, during the study a newer tracker they released gained popularity: the Chipolo POP. POP trackers work with either Apple Find My or Google Find My Device similar to the way the Pebblebee Clip 5 does. Due to their integration with these finding networks, they may be more DULT compliant than the older ONE trackers which work off of their own network.

6.2 Future Works

More work needs to be done to remedy the above limitations. If continuing with a policy-level evaluation, the web query collection needs to be vastly expanded so that a claim that most policy had

been collected could be validated. To this end, web scraping would most likely be necessary to increase scale and help reduce any possible sampling bias. To deal with an increase of data and to systematize the process, it would be more logical to employ some sort of code-book over the data.

To truly understand the DULT adherence landscape to the fullest extent, a study delving beyond policy-level knowledge is necessary. Prior work on Tile [22], SmartTag [42] and AirTag [1] reverse engineering could be referenced to evaluate the technical level specifications on the list, and reverse engineering could be done on the less popular manufacturers such as Pebblebee and Chipolo.

It may also be fruitful to reach out to manufacturers to see if they would be willing to provide limited technical specifications to researchers to validate the checklist, especially if they claim to adhere to DULT.

7 CONCLUSION

We provide the first policy-level evaluation of DULT adherence across major consumer Bluetooth trackers. Out results find that across all devices, publicly disclosed policies fail to meaningfully demonstrate DULT compliance. Additionally, some manufacturers explicitly disallow reverse engineering either in terms of service [13]. With policy-level documentation lacking depth alongside an inability for technologically savvy users to investigate compliance themselves without violating TOS, this leaves no legal way for a user to validate DULT adherence. This lack of transparency impedes third-party evaluation and limits manufacturer accountability.

Our results show that user safety cannot rely on policy disclosures alone. Although we note a correlated decrease in stalking attack vectors amongst devices which adhere to DULT practices, stalking attack vectors still remain feasible. Additionally, our findings suggest that theft attacks are trivial against devices which adhere to DULT.

Together, these findings highlight the need for stronger transparency requirements, auditable implementations, and mechanisms that allow for independent validation of safety-critical behaviors.

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8 APPENDIX

8.1 Full Query Table

Table 2: Web Search Queries and Results by Manufacturer

Manufacturer	Query	Result Link
Life360 Tile	Anti-stalking	https://www.life360.com/blog/tile-anti-theft-mode
Life360 Tile	Anti-stalking	https://support.life360.com/hc/en-us/articles/30582923425815-Tile-Anti-Theft-Mode
Life360 Tile	Anti-stalking	https://www.life360.com/blog/tile-introduces-scan-and-secure-feature-unwanted-tracking-safety
Life360 Tile	Unwanted tracking	https://support.life360.com/hc/en-us/articles/30582987898135-How-to-Handle-Unwanted-Tracking
Life360 Tile	Unwanted tracking	https://www.life360.com/blog/tile-introduces-scan-and-secure-feature-unwanted-tracking-safety
Life360 Tile	Unwanted tracking	https://support.life360.com/hc/en-us/articles/30582923425815-Tile-Anti-Theft-Mode
Life360 Tile	Anti-theft	https://support.life360.com/hc/en-us/articles/30582923425815-Tile-Anti-Theft-Mode
Life360 Tile	Anti-theft	https://www.life360.com/blog/tile-anti-theft-mode
Life360 Tile	Anti-theft	https://www.life360.com/blog/how-does-tile-anti-theft-mode-work
Life360 Tile	Privacy policy	https://life360-legal.zendesk.com/hc/en-us/articles/16038777217175-Life360-Privacy-Policy
Life360 Tile	Privacy policy	https://support.life360.com/hc/en-us/articles/30583010520087-Tile-Security
Life360 Tile	Privacy policy	https://life360-legal.zendesk.com/hc/en-us/articles/27124908196375-Tile-Privacy-Policy-previous-to-January-3-2023
Apple AirTag	Anti-stalking	https://www.apple.com/newsroom/2022/02/an-update-on-airtag-and-unwanted-tracking/
Apple AirTag	Anti-stalking	https://support.apple.com/en-us/119874
Apple AirTag	Anti-stalking	https://support.apple.com/guide/personal-safety/detect-unwanted-trackers-ips139b15fd9/web
Apple AirTag	Unwanted tracking	https://support.apple.com/en-us/119874
Apple AirTag	Unwanted tracking	https://support.apple.com/guide/personal-safety/detect-unwanted-trackers-ips139b15fd9/web
Apple AirTag	Unwanted tracking	https://www.apple.com/newsroom/2022/02/an-update-on-airtag-and-unwanted-tracking/
Apple AirTag	Anti-theft	https://support.apple.com/en-gb/119874
Apple AirTag	Anti-theft	https://support.apple.com/en-us/120340
Apple AirTag	Anti-theft	No further results (Google results page 10 reached)
Apple AirTag	Privacy policy	https://www.apple.com/legal/privacy/en-ww/
Apple AirTag	Privacy policy	https://www.apple.com/newsroom/2022/02/an-update-on-airtag-and-unwanted-tracking/
Apple AirTag	Privacy policy	https://www.apple.com/privacy/control/
Samsung SmartTag	Anti-stalking	https://www.samsung.com/us/support/answer/ANS00088244/
Samsung SmartTag	Anti-stalking	https://www.samsung.com/ae/support/mobile-devices/how-to-disable-smarttag-2-location-tracking/
Samsung SmartTag	Anti-stalking	https://news.samsung.com/us/samsung-smartthings-find-expands-over-300-million-nodes-helping-locate-devices/
Samsung SmartTag	Unwanted tracking	https://www.samsung.com/ae/support/mobile-devices/how-to-disable-smarttag-2-location-tracking/
Samsung SmartTag	Unwanted tracking	https://www.samsung.com/us/support/answer/ANS00088244/
Samsung SmartTag	Unwanted tracking	https://news.samsung.com/us/introducing-new-samsung-galaxy-smarttag2-a-smart-way-to-keep-track-of-important-things-in-your-life/

Manufacturer	Query	Result Link
Samsung SmartTag	Anti-theft	https://www.samsung.com/us/support/answer/ANS00088244/
Samsung SmartTag	Anti-theft	https://news.samsung.com/us/introducing-new-samsung-galaxy-smarttag2-a-smart-way-to-keep-track-of-important-things-in-your-life/
Samsung SmartTag	Anti-theft	https://news.samsung.com/global/introducing-new-samsung-galaxy-smarttag2-smarter-ways-to-keep-track-of-valuables
Samsung SmartTag	Privacy policy	https://www.samsung.com/us/account/privacy-policy/
Samsung SmartTag	Privacy policy	https://www.samsung.com/us/privacy/
Samsung SmartTag	Privacy policy	https://www.samsung.com/us/privacy/rights/
Chipolo One	Anti-stalking	https://chipolo.net/en-us/blogs/chipolo-one-spot-how-it-works
Chipolo One	Anti-stalking	https://support.chipolo.net/hc/en-us/articles/360021040938-Do-Chipolo-ONE-Spot-and-CARD-Spot-have-Unwanted-Tracking-Protection
Chipolo One	Anti-stalking	https://chipolo.net/en-us/products/category/chipolo-spot
Chipolo One	Unwanted tracking	https://support.chipolo.net/hc/en-us/articles/360021040938-Do-Chipolo-ONE-Spot-and-CARD-Spot-have-Unwanted-Tracking-Protection
Chipolo One	Unwanted tracking	https://chipolo.net/es/blogs/google-and-apple-cooperate-on-unwanted-bluetooth-tracking-alerts
Chipolo One	Unwanted tracking	https://chipolo.net/es/blogs/chipolo-one-spot-how-it-works
Chipolo One	Anti-theft	https://chipolo.net/en-us/blogs/chipolo-one-spot-how-it-works
Chipolo One	Anti-theft	https://chipolo.net/en-us/pages/chipolo-collection
Chipolo One	Anti-theft	https://chipolo.net/en-us/blogs/introducing-the-new-chipolo-one
Chipolo One	Privacy policy	https://chipolo.net/en-us/pages/privacy-policy
Chipolo One	Privacy policy	https://support.chipolo.net/hc/en-us/articles/360018406458-Sharing-what-does-it-mean-for-my-privacy
Chipolo One	Privacy policy	https://chipolo.net/en-us/pages/terms-of-use
Pebblebee Clip 5	Anti-stalking	https://pebblebee.com/blogs/news/pebblebee-pioneering-solutions-against-unwanted-tracking
Pebblebee Clip 5	Anti-stalking	https://pebblebee.com/pages/alert
Pebblebee Clip 5	Anti-stalking	https://pebblebee.com/pages/universal-clip-5-reasons
Pebblebee Clip 5	Unwanted tracking	https://help.pebblebee.com/en-US/unwanted-location-tracking-alerts-pebblebee-2536869
Pebblebee Clip 5	Unwanted tracking	https://pebblebee.com/blogs/news/pebblebee-pioneering-solutions-against-unwanted-tracking
Pebblebee Clip 5	Unwanted tracking	https://pebblebee.com/products/clip-5
Pebblebee Clip 5	Anti-theft	https://help.pebblebee.com/en-US/understanding-the-new-clip-5-3153344
Pebblebee Clip 5	Anti-theft	https://pebblebee.com/products/clip-5
Pebblebee Clip 5	Anti-theft	https://pebblebee.com/collections/bicycle-security
Pebblebee Clip 5	Privacy policy	https://pebblebee.com/pages/privacy-policy
Pebblebee Clip 5	Privacy policy	https://pebblebee.com/products/clip-5
Pebblebee Clip 5	Privacy policy	https://pebblebee.com/pages/faqs

8.2 Full List of Consulted Links

Table 3: Full list of consulted links

Links
https://www.life360.com/blog/tile-anti-theft-mode
https://support.life360.com/hc/en-us/articles/30582923425815-Tile-Anti-Theft-Mode
https://www.life360.com/blog/how-does-tile-anti-theft-mode-work
https://support.life360.com/hc/en-us/articles/30582996499095-Tile-Scan-and-Secure-Overview
https://support.life360.com/hc/en-us/articles/30582950641559-Tile-Scan-and-Secure-FAQ
https://www.life360.com/blog/tile-introduces-scan-and-secure-feature-unwanted-tracking-safety
https://support.life360.com/hc/en-us/articles/30582987898135-How-to-Handle-Unwanted-Tracking
https://life360-legal.zendesk.com/hc/en-us/articles/16038777217175-Life360-Privacy-Policy
https://www.apple.com/newsroom/2022/02/an-update-on-airtag-and-unwanted-tracking/
https://support.apple.com/en-us/119874
https://support.apple.com/guide/personal-safety/detect-unwanted-trackers-ips139b15fd9/web
https://www.apple.com/legal/privacy/en-ww/
https://www.apple.com/privacy/control/
https://support.apple.com/en-us/102170
https://www.samsung.com/us/support/answer/ANS00088244/
https://www.samsung.com/ae/support/mobile-devices/how-to-disable-smarttag-2-location-tracking/
https://news.samsung.com/us/samsung-smartthings-find-expands-over-300-million-nodes-helping-locate-devices/
https://news.samsung.com/us/introducing-new-samsung-galaxy-smarttag2-a-smart-way-to-keep-track-of-important-things-in-your-life/
https://news.samsung.com/global/introducing-new-samsung-galaxy-smarttag2-smarter-ways-to-keep-track-of-valuables
https://www.samsung.com/us/account/privacy-policy/
https://support.smartthings.com/hc/en-us/articles/10863369660052-SmartThings-Find
https://r1.community.samsung.com/t5/tips-tricks/activate-lost-mode-on-galaxy-smarttag
https://chipolo.net/en-us/blogs/chipolo-one-spot-how-it-works
https://chipolo.net/es/blogs/google-and-apple-cooperate-on-unwanted-bluetooth-tracking-alerts
https://chipolo.net/en-us/blogs/introducing-the-new-chipolo-one
https://chipolo.net/en/blogs/out-of-range-alerts-a-new-chipolo-feature?blog_introducing_the_new_chipolo_one
https://chipolo.net/en-us/products/chipolo-one
https://support.chipolo.net/hc/en-us/articles/360021040938-Do-Chipolo-ONE-Spot-and-CARD-Spot-have-Unwanted-Tracking-Protection
https://chipolo.net/en-us/pages/privacy-policy
https://support.chipolo.net/hc/en-us/articles/360018406458-Sharing-what-does-it-mean-for-my-privacy
https://support.chipolo.net/hc/en-us/articles/209527125-How-To-Share-Your-Chipolo
https://chipolo.net/en-us/pages/terms-of-use
https://support.chipolo.net/hc/en-us/articles/212204165-What-is-the-Chipolo-Community
https://support.chipolo.net/hc/en-us/articles/212175765-How-does-Chipolo-work
https://support.chipolo.net/hc/en-us/articles/212173825-I-found-someone-s-item-with-the-Chipolo-attached-what-should-I-do
https://support.chipolo.net/hc/en-us/sections/26588123194397-Maintenance-and-reset
https://pebblebee.com/blogs/news/pebblebee-pioneering-solutions-against-unwanted-tracking
https://pebblebee.com/pages/alert
https://pebblebee.com/pages/universal-clip-5-reasons
https://pebblebee.com/products/clip-5
https://help.pebblebee.com/en-US/understanding-the-new-clip-5-3153344
https://pebblebee.com/collections/bicycle-security
https://pebblebee.com/pages/privacy-policy
https://pebblebee.com/pages/faqs
https://support.google.com/android/answer/13658562?hl=en#zippy=%2Cyouve-got-an-unknown-tracker-alert%2Cplay-sound-to-find-an-unknown-tracker%2Cdisable-the-tracker

8.3 Full DULT Table

Table 4: DULT Best Practices Compliance Table

Practice	Tile	AirTag	SmartTag	Chipolo ONE	Pebblebee Clip 5
The accessory SHALL use Bluetooth Low Energy (LE) as the transport protocol.	Yes	Yes	Yes	Yes	Yes
The accessory SHALL advertise using Bluetooth LE.	Yes	Yes	Yes	Yes	Yes
The accessory MUST support at least one non-owner unencrypted connection in a peripheral role.	No evidence	No evidence	No evidence	No evidence	No evidence
Non-owner connections to the accessory SHALL be implemented using a platform-compatible method.	No evidence	Yes	Yes	Yes	Yes
The accessory SHALL maintain an internal state that detects when its location is, or has been, available to the owner via a network. This state is called the location-enabled state.	No evidence	No evidence	No evidence	No evidence	No evidence
When in location-enabled state, the accessory SHALL advertise a Bluetooth LE format recognizable to the platforms.	No evidence	Yes	No evidence	No evidence	No evidence
The advertisement MUST allow the owner's platform to reliably recognize the owner's associated accessories.	Yes	Yes	Yes	No evidence	Yes
The accessory SHALL NOT use its public address as its public identity when exchanging pairing keys at phase 3.	No evidence	No evidence	No evidence	No evidence	No evidence
It SHALL only use a static random address.	No evidence	No evidence	No evidence	No evidence	No evidence
The accessory SHALL broadcast the location-enabled advertisement payload if location is available to the owner or was available any time within the past 24 hours.	No evidence	No evidence	No evidence	No evidence	No evidence
The accessory SHALL transition from near-owner mode to separated mode if accessory has been physically separated from the owner device for more than 30 minutes	No evidence	No evidence	No evidence	No evidence	No evidence
The accessory SHALL transition from separated to near-owner mode if it has reunited with the owner device for a duration no longer than 30 minutes.	No evidence	No evidence	No evidence	No evidence	No evidence
The Bluetooth LE advertisement payload SHALL contain an address in the 6-byte MAC address field which looks random while recognizable to the owner device.	No evidence	No evidence	No evidence	No evidence	No evidence
The address SHALL rotate periodically.	No evidence	Yes	No evidence	No evidence	No evidence
It is possible to generate the MAC address in a way which meets the privacy requirement while allowing the platform to recognize an owned accessory without ambiguity using the MAC address, as defined in Section 6.1.1. When taking this approach, the address type SHALL be set as a non-resolvable private address or as a static device address.	No evidence	No evidence	No evidence	No evidence	No evidence
The owner MUST be able to predict the MAC address value at any given time in order to suppress unwanted tracking alerts caused by a device's owned accessory.	No evidence	Yes	No evidence	No evidence	No evidence
An accessory SHALL rotate its address on any transition between near-owner and separated states.	No evidence	No evidence	No evidence	No evidence	No evidence
When in near-owner state, the accessory SHALL rotate its address every 15 minutes.	No evidence	No evidence	Yes	No evidence	No evidence
When in separated state, the accessory SHALL rotate its address every 24 hours.	No evidence	No evidence	No evidence	No evidence	No evidence
The 1-byte Network ID SHALL be set based on a registered value for the manufacturer.	No evidence	No evidence	No evidence	No evidence	No evidence

Practice	Tile	AirTag	SmartTag	Chipolo ONE	Pebblebee Clip 5
The values of payload which may be different between accessories SHALL rotate at the same time and interval as the MAC address.	No evidence	No evidence	No evidence	No evidence	No evidence
If a Resolvable Private MAC address is used, this field SHALL be populated with a value of 6 bytes minimum.	No evidence	No evidence	No evidence	No evidence	No evidence
The accessory MUST set the near-owner bit to be 1 when in near-owner mode, otherwise 0.	No evidence	No evidence	No evidence	No evidence	No evidence
The near-owner bit MUST be the least significant bit.	No evidence	No evidence	No evidence	No evidence	No evidence
A maximum advertising interval of 4 seconds SHALL be used.	No evidence	No evidence	No evidence	No evidence	No evidence
The accessory non-owner service UUID SHALL be 15190001-12F4-C226-88ED-2AC5579F2A85.	No evidence	No evidence	No evidence	No evidence	No evidence
This service SHALL use GATT over LE.	No evidence	No evidence	No evidence	No evidence	No evidence
The service SHALL be instantiated as a primary service.	No evidence	No evidence	No evidence	No evidence	No evidence
The accessory non-owner characteristic UUID SHALL be 8E0C0001-1D68-FB92-BF61-48377421680E.	No evidence	No evidence	No evidence	No evidence	No evidence
The characteristic SHALL be transmitted little-endian.	No evidence	No evidence	No evidence	No evidence	No evidence
The accessory SHALL NOT request an MTU smaller than the maximum length of its write responses. In other words, all opcode response data must fit within a single write operation.	No evidence	No evidence	No evidence	No evidence	No evidence
The following accessory information MUST be persistent through the lifetime of the accessory: Product data (Section 3.12.1.1), Manufacturer name (Section 3.12.1.2), Model name (Section 3.12.1.3), Accessory category (Section 3.12.1.4), Accessory capabilities (Section 3.12.1.6), Network ID (Section 3.7), and Battery Type (Section 3.12.1.9).	No evidence	No evidence	No evidence	No evidence	No evidence
These opcodes SHALL be available when the accessory is in separated state. (see DULT table 4)	No evidence	No evidence	No evidence	No evidence	No evidence
These opcodes SHALL NOT be available when the accessory is in near-owner state. (see DULT table 4)	No evidence	No evidence	No evidence	No evidence	No evidence
When any opcode is not available, the accessory SHALL return the Invalid_command error as the ResponseStatus in Command_Response.	No evidence	No evidence	No evidence	No evidence	No evidence
If an optional opcode is not available, the accessory SHALL return the Invalid_command error as the ResponseStatus in Command_Response.	No evidence	No evidence	No evidence	No evidence	No evidence
If any opcode value is commanded that is not supported by the accessory, it SHALL return the Invalid_command error as the ResponseStatus in the Command_Response.	No evidence	No evidence	No evidence	No evidence	No evidence
In the circumstances that there are multiple non-owner connections, all GATT indication subprocedures defined in Table 4 SHALL be sent through only to the connection that commanded the affiliated write subprocedure.	No evidence	No evidence	No evidence	No evidence	No evidence
The Product Data operand represents an 8-byte value that is intended to serve as a unique identifier for the accessory make and model. This value SHALL be determined during the onboarding process	No evidence	No evidence	No evidence	No evidence	No evidence
When the Manufacturer Name is less than 64 bytes, it SHALL be formatted either as: a string value with length less than 64 bytes Or a string value that is both zero-terminated and zero-padded up to 64 bytes	No evidence	No evidence	No evidence	No evidence	No evidence

Practice	Tile	AirTag	SmartTag	Chipolo ONE	Pebblebee Clip 5
The firmware revision string SHALL use the x[.y[.z]] format where : <x>is the major version number, required. <y>is the minor version number, required if it is non zero or if <z>is present. <z>is the revision version number, required if non zero.	No evidence	No evidence	No evidence	No evidence	No evidence
The firmware revision MUST follow these rules: <x>is incremented when there is significant change; for example, 1.0.0, 2.0.0, 3.0.0, and so on. <y>is incremented when minor changes are introduced, such as 1.1.0, 2.1.0, 3.1.0, and so on. <z>is incremented when bug fixes are introduced, such as 1.0.1, 2.0.1, 3.0.1, and so on. Subsequent firmware updates can have a lower <y>version only if <x>is incremented. Subsequent firmware updates can have a lower <z>version only if <x>or <y>is incremented.	No evidence	No evidence	No evidence	No evidence	No evidence
(firmware) Major version MUST not be greater than (2^{16}-1.).	No evidence	No evidence	No evidence	No evidence	No evidence
(firmware) Minor and revision versions MUST not be greater than (2^{8}-1.).	No evidence	No evidence	No evidence	No evidence	No evidence
(firmware) The value MUST change after every firmware update.	No evidence	No evidence	No evidence	No evidence	No evidence
The network revision string SHALL use the x[.y[.z]] format where : <x>is the major version number, required. <y>is the minor version number, required if it is non zero or if <z>is present. <z>is the revision version number, required if non zero.	No evidence	Yes	No evidence	No evidence	No evidence
The network revision MUST follow these rules: <x>is incremented when there is significant change; for example, 1.0.0, 2.0.0, 3.0.0, and so on. <y>is incremented when minor changes are introduced, such as 1.1.0, 2.1.0, 3.1.0, and so on. <z>is incremented when bug fixes are introduced, such as 1.0.1, 2.0.1, 3.0.1, and so on. Subsequent network updates can have a lower <y>version only if <x>is incremented. Subsequent network updates can have a lower <z>version only if <x>or <y>is incremented.	No evidence	No evidence	No evidence	No evidence	No evidence
(network) Major version MUST not be greater than (2^{16}-1.).	No evidence	No evidence	No evidence	No evidence	No evidence
(network) Minor and revision versions MUST not be greater than (2^8 1).	No evidence	No evidence	No evidence	No evidence	No evidence
(network) The value MUST change after every network update.	No evidence	No evidence	No evidence	No evidence	No evidence
Once a user has been notified of an unknown accessory traveling with them„ it is REQUIRED that users have the means to physically locate the accessory.	Yes	Yes	Yes	No evidence	Yes
Includes Accelerometer	No evidence	Yes	Yes	No evidence	No evidence
If the accessory includes an accelerometer, it MUST detect ±10° orientation change.	No evidence	No evidence	No evidence	No evidence	No evidence
After TSEPARATED_UT_TIMEOUT in separated state, the accessory MUST enable the motion detector to detect any motion within TSEPARATED_UT_SAMPLING_RATE1.	No evidence	No evidence	No evidence	No evidence	No evidence
If motion is not detected within the TSEPARATED_UT_SAMPLING_RATE1 period, the accessory MUST stay in this state until it exits separated state.	No evidence	No evidence	No evidence	No evidence	No evidence

Practice	Tile	AirTag	SmartTag	Chipolo ONE	Pebblebee Clip 5
If motion is detected within the TSEPARATED_UT_SAMPLING_RATE1 the accessory MUST play a sound.	No evidence	No evidence	No evidence	No evidence	No evidence
The accessory MUST continue to play a sound for every detected motion.	No evidence	Yes	No evidence	No evidence	No evidence
The accessory SHALL disable the motion detector for TSEPARATED_UT_BACKOFF under either of the following conditions: Motion has been detected for 20 seconds at TSEPARATED_UT_SAMPLING_RATE2 periods. Ten sounds are played.	No evidence	No evidence	No evidence	No evidence	No evidence
If the accessory is still in separated state at the end of TSEPARATED_UT_BACKOFF, the UT behavior MUST restart.	No evidence	No evidence	No evidence	No evidence	No evidence
A Bluetooth LE connection from an associated device MUST reset the separated behavior	No evidence	No evidence	No evidence	No evidence	No evidence
The accessory MUST include a sound maker.	Yes	Yes	Yes	Yes	Yes
It MUST also play sound when a non-owner tries to locate the accessory by initiating a play sound command from a non-owner device when the accessory is in range and connectable through Bluetooth LE.	No	Yes	No evidence	No evidence	Yes
The sound maker MUST emit at least 60 phon peak loudness.	No evidence	No evidence	No evidence	No evidence	No evidence
The loudness MUST be measured in free acoustic space substantially free of obstacles that would affect the pressure measurement.	No evidence	No evidence	No evidence	No evidence	No evidence
The loudness MUST be measured by a calibrated (to the Pascal) free field microphone 25 cm from the accessory suspended in free space.	No evidence	No evidence	No evidence	No evidence	No evidence
Non-owner controls SHALL use the same UUIDs as Accessory Connections.	No evidence	No evidence	No evidence	No evidence	No evidence
Sound_Start and Sound_Stop SHALL only be available in separated state.	No evidence	No evidence	No evidence	No evidence	No evidence
In all other states, the accessory SHALL return Invalid_command.	No evidence	No evidence	No evidence	No evidence	No evidence
If Identifer Retrieval over Bluetooth LE (Section 3.15.3) is supported, Get_Identifier SHALL only be available when in identifier read state; otherwise, it MUST send Command_Response (Section 3.13.4.1.1) with the Invalid_command as the ResponseStatus.	No evidence	No evidence	No evidence	No evidence	No evidence
In the circumstances that there are multiple non-owner connections, all GATT indication subprocedures defined in Table 18 SHALL be sent through only to the connection that commanded the affiliated write subprocedure	No evidence	No evidence	No evidence	No evidence	No evidence
The sound maker MUST play sound for 5–30 seconds.	No evidence	No evidence	No evidence	No evidence	No evidence
The accessory SHALL confirm start via Command_Response.	No evidence	No evidence	No evidence	No evidence	No evidence
The accessory SHALL confirm completion with Sound_Completed.	No evidence	No evidence	No evidence	No evidence	No evidence
The accessory SHALL respond to any invalid opcode with Command_Response and Invalid_command as the ResponseStatus	No evidence	No evidence	No evidence	No evidence	No evidence
The Get_Identifier opcode is used to retrieve identifier lookup payload over Bluetooth LE. To enable this opcode, the accessory MUST be in the identifier read state.	No evidence	No evidence	No evidence	No evidence	No evidence

Practice	Tile	AirTag	SmartTag	Chipolo ONE	Pebblebee Clip 5
To enter the identifier read state, a user action on the accessory MUST be performed (for example, press and hold a button for 10 seconds).	No evidence	Yes	No evidence	No evidence	No evidence
he identifier read state MUST be enabled for 5 minutes once the user action on the accessory is successfully performed.	No evidence	No evidence	No evidence	No evidence	No evidence
When the accessory is in this mode, it MUST respond with Get_Identifier_Response opcode and Identifier Payload operand.	No evidence	No evidence	No evidence	No evidence	No evidence
It is REQUIRED that the encrypted identifier (which in some cases is the product serial number) be non-identifiable.	No evidence	No evidence	No evidence	No evidence	No evidence
If the accessory is not in identifier read state, it MUST send Command_Response (Section 3.13.4.1.1) with the Invalid_command as the ResponseStatus	No evidence	No evidence	No evidence	No evidence	No evidence
The accessory SHALL have a way to be disabled such that locations cannot be seen.	Yes	Yes	Yes	Yes	Yes
Disablement SHALL be done via some physical action (e.g., button press, gesture, removal of battery, etc.).	No	Yes	Yes	Yes	Yes
The accessory manufacturer SHALL provide both a text description of how to disable the accessory as well as a visual depiction (e.g. image, diagram, animation, etc.) that MUST be available when the platform is online and OPTIONALLY when offline.	No evidence	Yes	No evidence	No	No evidence
The accessory MUST include a way to uniquely identify it - either via a serial number or other privacy-preserving solution.	No evidence	Yes	Yes	Yes	Yes
If a serial number is available, it SHALL be printed and easily accessible.	No evidence	Yes	No evidence	No evidence	No evidence
The serial number MUST be unique for each product ID.	No evidence	No evidence	No evidence	No evidence	No evidence
Identifier payload SHALL be readable via NFC or BLE.	No evidence	Yes	Yes	No evidence	No evidence
For privacy reasons, accessories that support identifier retrieval for identifiers not included in the advertising packet over Bluetooth LE MUST have a physical mechanism, for example, a button, that SHALL be required to enable the Get_Identifier opcode	No evidence	No evidence	No evidence	No evidence	No evidence
Identifier payload returned in paired state SHALL be encrypted.	No evidence	No evidence	No evidence	No evidence	No evidence
For those accessories that support identifier retrieval over NFC, an associated accessory SHALL advertise the whole URL with arguments as the payload over NFC.	No evidence	No evidence	No evidence	No evidence	No evidence
The URL SHALL be hosted by the network provider.	No evidence	No evidence	No evidence	No evidence	No evidence
The URL SHALL decrypt the identifier payload and return the identifier of the accessory in a form that can be rendered in the platform's HTML view.	No evidence	No evidence	No evidence	No evidence	No evidence
When a user performs NFC Tap and the accessory is in associated state, the encrypted identifier encoded in hex string SHALL be an argument ("e") passed to the identifier retrieval URL.	No evidence	No evidence	No evidence	No evidence	No evidence
Verifiable identity information of the owner of an accessory at time of association SHALL be recorded and associated with the identifier of the accessory, e.g., phone number, email address.	No evidence	Yes	Yes	No evidence	No evidence

Practice	Tile	AirTag	SmartTag	Chipolo ONE	Pebblebee Clip 5
This information SHALL be part of the response of the identifier retrieval from a server which can be rendered in a platform's HTML view.	No evidence	No evidence	No evidence	No evidence	No evidence
Obfuscated owner information SHALL be made available. This MUST include at least one of: last four digits of phone number or partially obfuscated email.	No	Yes	No evidence	No evidence	Yes
Includes NFC	No	Yes	Yes	No	No
The payload for NFC tags MUST use NDEF URI Record Type Definition as defined by NFC Forum™ in URI Record Type Definition RTD-URI 1.0 NFCForum-TS-RTD URI 1.0.		No evidence	No evidence		
NFC tags MUST use the NFC Data Exchange Format (NDEF) as defined by NFC Forum™ in NDEF 1.0 NFCForum-TS-NDEF 1.0.		No evidence	No evidence		
NFC tag types MUST be type 2 or greater.		No evidence	No evidence		
The NFC tag SHALL not be scannable when the accessory is still in the packaging.		No evidence	No evidence		
The payload MUST be scannable when holding an NFC-enabled device near the center of the NFC tag on the accessory.		Yes	Yes		
The NFC implemention of the accessory MUST be configured as a NFC tag.		Yes	Yes		
Accessory manufacturer's MUST pick an accessory category value that closest resembles their physical product. If none of the accessory categories provided in Table 23 match the physical product, Other MUST be chosen.	No evidence	No evidence	No evidence	No evidence	No evidence
Accessory manufacturers SHALL provide this set of recognizable values to the platform, along with an offset and length indicating what part of the advertisement to match. This set MUST be large enough to accommodate a time offset between the accessory and owner's host platform.	No evidence	No evidence	No evidence	No evidence	No evidence
Network ID SHALL be used to associate an accessory.	No evidence	No evidence	No evidence	No evidence	No evidence
Network access MUST NOT be required for owned accessory recognition.	No evidence	No evidence	No evidence	No evidence	No evidence
The platform MUST delete any local identifying information when the software is removed or unassociated.	No evidence	No evidence	No evidence	No evidence	No evidence
Accessory manufacturers MUST follow a minimum set of steps for their accessories to be detectable by platforms such as adding their Network ID value to the Finding Network Registry (Section 10.1).	No evidence	Yes	No evidence	No evidence	No evidence
During onboarding, a product data registry SHALL be created and maintained by the network provider for all accessory manufacturers participating in their network.	No evidence	No evidence	No evidence	No evidence	No evidence
Obfuscated owner information SHALL only be available in separated mode.	No evidence	No evidence	No evidence	No evidence	No evidence
When requested through any long range wireless interface like Bluetooth, a user action MUST be required for the requesting device to access the obfuscated owner information and identifier.	No evidence	No evidence	No evidence	No evidence	No evidence
To uphold privacy and anti-tracking features like the Bluetooth MAC address randomization, the accessory MUST only provide non-identifiable data to non-owner requesting devices.	No evidence	No evidence	No evidence	No evidence	No evidence